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Categories: Foundational Definition and Examples

What is a Category?

A **category** is an abstract structure capturing: - **Objects**: Things being studied (sets, groups, topological spaces, etc.) - **Morphisms**: Structure-preserving maps between objects - **Composition**: How morphisms combine

Formal Definition

Category $C = (\text{Ob}(C), \text{Mor}(C), \circ, \text{id})$ where:

1. Collections: - $\text{Ob}(C)$: objects of C - For each $X, Y \in \text{Ob}(C)$: $\text{Hom}_C(X, Y)$ = morphisms $f: X \rightarrow Y$

2. Composition:

$$\circ: \text{Hom}(Y, Z) \times \text{Hom}(X, Y) \rightarrow \text{Hom}(X, Z)$$

$$(g, f) \mapsto g \circ f$$

3. Identities:

For each $X \in \text{Ob}(C)$: $\text{id}_X \in \text{Hom}(X, X)$

4. Axioms:

ASSOCIATIVITY: $(h \circ g) \circ f = h \circ (g \circ f)$

IDENTITY: $\text{id}_Y \circ f = f = f \circ \text{id}_X$

11 Concrete Examples**1. Set: Sets and Functions**

Objects: All sets Morphisms: Functions $f: X \rightarrow Y$ Composition: Function composition Identity: $\text{id}_X(x) = x$

2. Top: Topological Spaces

Objects: Topological spaces Morphisms: Continuous functions Composition: Functional composition
(continuous $\circ$ continuous = continuous)

3. Grp: Groups and Homomorphisms

Objects: Groups $(G, \cdot)$ Morphisms: Group homomorphisms $\phi: G \rightarrow H$ (preserve multiplication) Composition: Function composition (homomorphism $\circ$ homomorphism = homomorphism)

4. Vect_k: Vector Spaces

Objects: Vector spaces over field k Morphisms: k -linear maps $f: V \rightarrow W$ ($f(av + bw) = af(v) + bf(w)$)

5. Ring: Rings and Ring Homomorphisms

Objects: Rings $(R, +, \times)$ Morphisms: Ring homomorphisms (preserve both operations)

6. Poset: Partially Ordered Sets

Objects: Posets $(S, \leq)$ Morphisms: $f: S \rightarrow T$ with $x \leq y \implies f(x) \leq f(y)$ (order-preserving)

7. Rel: Relations

Objects: Sets Morphisms: Relations $R \subseteq X \times Y$ Composition: Relational composition $\{(x,z) \mid \exists y: (x,y) \in R, (y,z) \in S\}$

8. Mon: Monoids

Objects: Monoids $(M, \cdot, e)$ Morphisms: Monoid homomorphisms

9. Cat: The Category of Categories

Objects: Small categories Morphisms: Functors (structure-preserving maps between categories)

10. Met: Metric Spaces

Objects: Metric spaces (X, d) Morphisms: Continuous functions (distance-preserving)

11. Measured Spaces

Objects: Measurable spaces (X, Σ) Morphisms: Measurable functions

Morphism Properties

Isomorphism

$f: X \rightarrow Y$ with $\exists g: Y \rightarrow X$ where $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$

Interpretation: X and Y are “same” in category

Monomorphism (Monic)

$f: X \rightarrow Y$ such that $f \circ g_1 = f \circ g_2 \implies g_1 = g_2$

In Set: Injective functions **General:** Left-cancellative

Epimorphism (Epic)

$f: X \rightarrow Y$ such that $h_1 \circ f = h_2 \circ f \implies h_1 = h_2$

In Set: Surjective functions **General:** Right-cancellative **Warning:** NOT always surjective! (Example: $\mathbb{Z} \rightarrow \mathbb{Q}$ in Ring)

Endomorphism

$f: X \rightarrow X$ (morphism from object to itself)

Automorphism

$f: X \rightarrow X$ that is isomorphism

Bimorphism

f that is both monic and epic (but may not be iso!)

Commutative Diagrams

Visual representation of morphism relationships:

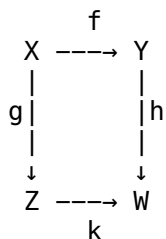


Diagram commutes if: $k \circ g = h \circ f$

Small vs Large Categories

Locally Small: $\text{Hom}_C(X,Y)$ is a set (not proper class)

Small Category: $\text{Ob}(C)$ and $\text{Mor}(C)$ are sets

Large Category: Allow proper classes of objects/morphisms Example: Category Cat of all categories (proper class of objects)

Why Categories?

1. **Unifies Mathematics:** Same structure in algebra, topology, logic
2. **Universal Properties:** Characterize objects by relationships
3. **Morphisms Over Elements:** Focus on structure-preserving maps
4. **Duality:** Opposite category gives free theorems
5. **Abstraction:** Common patterns across disciplines

Common Mistakes

- ✗ “Objects are primary, morphisms secondary” ✓ Actually: Morphisms define structure; objects are just labels
- ✗ “Monomorphism = Injective”
✓ Only in concrete categories; $\mathbb{Z} \rightarrow \mathbb{Q}$ is monic epic but not iso
- ✗ “Categories require elements” ✓ No: Categories are purely structural

Key Takeaways

- Categories capture mathematical structure abstractly
- Morphisms are the primary data
- Composition respects structure
- Many familiar structures are categories
- Universal properties are morphism conditions

Study Questions

1. What makes a morphism an isomorphism?
2. Why isn't every bimorphism an isomorphism?
3. How do limits generalize universal properties?
4. What does locally small mean?
5. Why focus on morphisms instead of elements?